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DR. B.V. HIRAY COLLEGE OF MANAGEMENT AND RESEARCH CENTRE

Malegaon Camp, Malegaon — Affiliated to Savitribai Phule Pune University, Pune (SPPU)

Approved by AICTE, New Delhi · NAAC Accredited

MINI PROJECT REPORT

CAMPUS SHIKAYAT SETU

College Grievance Redressal System

Full-stack web application for complaint lodging, automatic department routing, SLA-bound resolution and print-ready administrative registers

Course: Application Development (Mini Project — Semester I)

Program: Master of Computer Applications (MCA) · Sem-I

Academic Year: 2026–27

Submitted by

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Under the guidance of

Mrs. Priyanka M. Jagtap — Project Guide

CERTIFICATE

This is to certify that the Mini Project entitled “CAMPUS SHIKAYAT SETU — College Grievance Redressal System” is a bona fide work carried out by Mr. Yashwant Santosh Ahire, student of MCA-I (Semester-I), Dr. B.V. Hiray College of Management and Research Centre, Malegaon, under the SPPU MCA curriculum (Course: Application Development), during the academic year 2026–27.

The project has been completed satisfactorily and is submitted in partial fulfilment of the requirements of the Mini Project Review schedule issued by the Department of MCA (Review 3 on 21-11-2026 and the Final Internal Viva on 19-12-2026).

To the best of our knowledge, the matter embodied in this report is original and has not been submitted to any other University or Institution for the award of any degree or diploma.

Date: _____

Place: Malegaon

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Mrs. Priyanka M. Jagtap

Project Guide

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INDEX

Certificate	**2**
Chapter 1 — Introduction	**4**
Chapter 2 — Proposed System	**6**
Chapter 3 — System Analysis and Design	**8**
Chapter 4 — Coding	**13**
Chapter 5 — Testing	**15**
Chapter 6 — Limitations of Proposed System	**16**
Chapter 7 — Proposed Enhancements	**17**
Chapter 8 — Conclusion	**18**
Chapter 9 — Bibliography	**19**
Chapter 10 — User Manual	**20**
Appendix A — Database Schema (schema.sql)	**23**
Appendix B — Seed Data (actual records, ≥ 5 per table)	**25**
Appendix C — Glossary and Acronyms	**27**

List of Figures and Tables is maintained in-line within each chapter. All “screens” reproduced in this report are faithful renderings of the running application captured from the live build, not photographs.

CHAPTER 1

Introduction

1.1 Abstract

The college of Dr. B.V. Hiray College of Management and Research Centre, Malegaon, receives a steady stream of student grievances — leaking taps, dead projectors, missing marks, mess-food and security issues. Today they are lodged in a paper complaint box: they vanish into a pile, are routed by hand (if at all), carry no deadline and produce no management information. **Campus Shikayat Setu** is a full-stack web application that digitises the entire grievance workflow: lodging with category and location, **automatic department routing** via a keyword-weighted engine, an **SLA matrix** (High 24 h, Medium 72 h, Low 7 days) with computed overdue flags, officer workflow (acknowledge → in-progress → resolve), student feedback with a 7-day reopen window, anonymous complaints, and six print-ready administrative registers.

The system is built with **Python + Flask**, **Jinja2** templates, **vanilla JavaScript** and a **SQLite 3** database chosen deliberately for the use case (single campus, single PC, ≤ ~2,000 users; one-file backup; schema maps 1:1 to MySQL 8 for production). Three roles are supported — **student**, **department officer** and **administrator** — with strict server-side authorisation on every route.

Keywords: Grievance Redressal, Web Application, Flask, SQLite, SLA, Keyword Routing, SPPU MCA, Application Development.

1.2 Existing System and Need for System

1.2.1 Description of the existing (manual) system

Grievances are written on slips and dropped into the complaint box near the administrative office. The office staff forwards slips by hand to a department “when convenient”. Status is never communicated to the complainant; repeat complaints are invisible; there is no register of what was raised, who owned it, or whether it was ever resolved.

1.2.2 Problems with the existing system

- **No routing logic:** a complaint about a wifi router may sit with the hostel office for weeks.
- **No deadline:** nothing is tracked as “overdue”; there is no accountability and no escalation.
- **No status to the student:** the complainant can never tell whether the issue was seen, assigned or fixed.
- **No anonymity:** sensitive complaints (ragging, canteen hygiene) are either not reported or reported to the wrong person.
- **No data:** the college cannot answer “Which department resolves slowest?” or “How many complaints this month?”.
- **No audit:** there is no record of who touched a complaint and when.

1.2.3 Why a new system is needed

- Give every grievance a code (CMP-YYYY-NNNN), an owner department, a priority and an SLA deadline at the moment of lodging.
- Remove manual routing with a transparent, keyword-weighted router (score printed on the complaint timeline).
- Close the loop with student feedback and a controlled reopen window.
- Produce print-ready registers for review meetings — pendency, SLA compliance, resolution time, trend, feedback, routing audit.
- Protect sensitive complainants with anonymous lodging (identity visible only to administrators).

1.3 Scope of the System

In scope

- Student self-registration (roll no./e-mail/mobile validation), login, logout, password change.
- Complaint lodging with category, description, location and optional anonymous flag.
- Automatic department routing with weighted keyword scoring + audit event; admin reassign with full trace.
- SLA matrix (High 24 h / Medium 72 h / Low 7 days) with live overdue computation and escalation-to-High.
- Officer workflow: acknowledge → in-progress → resolve (remark mandatory) · student confirmation with 1–5 star rating or reopen (≤ 7 days).
- Admin console: filter by department/status/priority, reassign, escalate, force-close stale resolved

complaints, manage officer accounts.

• Six print-ready registers: Department Pendency · SLA Compliance · Resolution Time · Monthly Trend · Feedback Summary · Routing Audit.

• Public tracker (complaint code + roll no., no login) and seeded deterministic demo (32 complaints across 6 departments).

Out of scope (deferred — see Chapter 7)

- E-mail / SMS / WhatsApp notifications; push mobile app; multi-college tenancy.
- Photo evidence upload; ERP integration; public grievance board.
- Machine-learning auto-classification (current router is deliberately deterministic and explainable).

1.4 Operating Environment — Hardware and Software

Hardware (minimum specification to run the project)

Component	Minimum	Recommended (dev/demo)
Processor	Intel Core i3 (any generation)	Intel Core i5 / Ryzen 5
RAM	2 GB	8 GB
Storage	150 MB free	1 GB free (SSD)
Display	1366 × 768	1920 × 1080
Network	None (single-machine)	LAN (demo from another PC)

Software

Software	Version used	Purpose
Operating System	GNU/Linux (Debian 12) · Windows 10/11	Development & demo environment
Python	3.11.2	Application runtime
Flask	3.x	Web framework (routing, sessions, flash)
Jinja2	3.x	Server-side templating
SQLite	3.x (file: shikayat_setu.db)	Relational database (schema maps to MySQL 8)
Werkzeug	3.x	WSGI stack + PBKDF2 password hashing
pytest	9.x	Unit / integration test runner (12 tests)
Browser	Chrome 120+ / Firefox 120+ / Edge 120+	Client
IDE / Editor	VS Code + Git + GitHub	Development & version control

1.5 Brief Description of Technology Used

Python 3 is the implementation language — readable, rapid to build with, and immediately familiar to the review panel. **Flask** provides routing, sessions and Jinja2 rendering without imposing an ORM; all SQL is explicit, bound-parameterised and visible in the source for examination. **Jinja2** templates keep presentation out of logic; the front end uses semantic HTML5, a hand-written navy/gold CSS theme and zero build tooling.

SQLite 3 was selected after matching databases to the use case: single-campus, low-write, read-mostly, one-machine demo with zero DBA. It enforces integrity with UNIQUE, CHECK and FOREIGN KEY constraints. The schema is intentionally portable — the production path is MySQL 8 with a one-file connection swap (see Chapter 7).

Security: passwords are PBKDF2-hashed (Werkzeug); every authenticated route checks the session role server-side (students cannot open the admin console — covered by test_11); SQL uses bound parameters; anonymous complaints hide identity from officers (admin-only visibility); status changes write immutable audit events.

1.5.1 Operating systems used

Development, seeding and all 12 automated tests were executed on GNU/Linux (Python 3.11.2). No OS-specific APIs are used; the app is runnable unchanged on Windows 10/11 lab machines via python run.py.

CHAPTER 2

Proposed System

2.1 Study of Similar Systems

Similar grievance tooling was studied before design: CPGRAMS (Government of India public-grievance portal), the academic complaint web apps found on GitHub (e.g. the “Smart Complaint Management System” concept), and the college’s own manual box. The design goal was to keep the idea but make the engineering real.

System	Type / Cost	Strengths	Limitations vs need
Manual complaint box (as-is)	Paper · Rs. 0	Zero cost, anonymous by accident	No routing, no status, no SLA, no reports
CPGRAMS (GoI)	Citizen portal · free	National scale, escalation matrix	Heavyweight; external identity; no college-department model
Smart Complaint Management System (GitHub reference)	Concept repo · free	Keyword routing + status tracking idea	Empty PHP backend (0 bytes), localStorage-only demo, no DB, no auth, no reports, no tests
College ERP grievance module	Paid module	Integrated with admissions ERP	Vendor lock-in, ₹-scale cost, slow change cycle, no SLA ownership model

2.2 Proposed System Overview

Campus Shikayat Setu keeps the proven workflow of the reference concept — lodge → auto-route → track — and replaces every placeholder with production-grade engineering: a real relational schema with foreign-key enforcement, PBKDF2 authentication with three roles, a transparent router whose scoring is printed on the complaint timeline, an SLA engine, a 7-day student reopen window, full audit events and six print registers.

- **Student layer:** register → lodge (category, location, anonymous flag) → track (code or roll no.) → confirm or reopen with feedback.
- **Officer layer:** department queue ordered by priority then recency; acknowledge → start work → resolve with mandatory remark.
- **Admin layer:** college-wide console — filter, reassign, escalate (priority→High), force-close stale resolved complaints, manage officer accounts.
- **Report layer:** six Ctrl+P print-ready registers, each with ≥ 5 live records from the seed.

2.3 Objectives

- O1 — Digitise grievance lodging with validation and a permanent, coded identity (CMP-YYYY-NNNN).
- O2 — Route every complaint automatically with an explainable keyword-weighted engine.
- O3 — Stamp an SLA deadline at submission; surface overdue items everywhere.
- O4 — Enforce a complete, audited lifecycle (Submitted → Acknowledged → In Progress → Resolved → Closed).
- O5 — Protect sensitive complainants via anonymous lodging with admin-only identity.
- O6 — Close the loop: student feedback (1–5 stars) and a controlled 7-day reopen window.
- O7 — Produce management registers (pendency, SLA compliance, resolution time, trend, feedback, routing audit).
- O8 — Deliver a deterministic, self-seeding demo runnable with one command on any lab PC.

2.4 Feasibility Study

Dimension	Assessment
Technical	Flask + SQLite requires only Python 3.11; no drivers, no services, no network. All libraries are pip-installable offline from PyPI.
Operational	Officers already acknowledge grievances informally; this formalises the same gesture with buttons. Training = one page (Chapter 10).
Economic	Rs. 0 licence cost; runs on any existing college PC; replacement of a register with a file backup.
Schedule	Design, build, seed, tests and documentation completed inside the Sem-I window; Review 3 demo on 21-11-2026.
Legal / Ethical	Anonymous complaints handled with admin-only identity access; all actions audit-logged; student data minimised to roll no., phone, e-mail.

2.5 Modules Identified

Module	Responsibility
Auth & Profile	Login (username or roll no.), logout, student self-registration, change password, account activation/deactivation.
Lodging	Category picker, validation (title ≥ 8 chars, description ≥ 20), anonymous flag, preview of routing score.
Routing Engine (routing.py)	Weighted keyword scoring (multi-word keywords weigh more; title counts double); safety words force High; mass-impact phrases bump one level; unmatched → General & Administration.
SLA & Escalation	Deadline = submission + SLA(priority); overdue computed at render time; admin escalation raises priority to High with an audit event.
Officer Workflow	Department-scoped queue (priority, then recency); acknowledge / start work / resolve with mandatory ≥ 10-char remark; timeline remarks.
Student Closure	Feedback (1–5 + comment) closes; or reopen (≤ 7 days, reason mandatory) returns to In Progress.
Admin Console	Filters (department/status/priority), reassign with reason, force-close after 7-day no-response, officer creation & activation.
Reports	Six print registers incl. SLA compliance % and per-category average resolution hours.

CHAPTER 3

System Analysis and Design

3.1 Requirement Specification (summary)

3.1.1 Functional requirements

ID	Requirement	Priority
FR-1	Student shall register with roll no., e-mail, mobile (Indian format) and password; duplicate roll nos. are rejected.	Must
FR-2	Any user shall be able to lodge a complaint (students), with title ≥ 8 chars and description ≥ 20 chars.	Must
FR-3	System shall auto-route on submit with weighted keyword scoring; unmatched → General & Administration.	Must
FR-4	System shall compute SLA due time at submission: High 24 h, Medium 72 h, Low 168 h.	Must
FR-5	Officers shall acknowledge / mark in-progress / resolve; resolution requires a ≥ 10-char remark.	Must
FR-6	Students shall close via feedback (1–5 stars) or reopen within 7 days of resolution.	Must
FR-7	Anonymous complaints shall hide complainant identity from all officers; visible to admins only.	Must
FR-8	Admin shall filter, reassign, escalate (→High) and force-close (7-day no-response).	Must
FR-9	Every state change shall append an immutable timeline event.	Must
FR-10	Officer queue shall show only the officer's department; students see only their own complaints.	Must
FR-11	System shall render six print registers with live SQL totals.	Should
FR-12	Public tracking shall require complaint code + roll no.	Should

3.1.2 Non-functional requirements

ID	Requirement
NFR-1	Login response < 300 ms on a lab PC; page render < 500 ms; no JS framework dependency.
NFR-2	Zero data loss on crash: SQLite ACID with foreign_keys ON; backup = file copy.
NFR-3	Passwords PBKDF2-hashed; no plaintext anywhere; sessions are signed cookies.
NFR-4	Print CSS hides chrome; registers fit A4 portrait without truncation.
NFR-5	Deterministic seed: fresh clone reproduces the exact demo dataset (< 200 ms).
NFR-6	Code style: PEP 8, bound parameters only, no string-concatenated SQL.

3.2 Entity–Relationship Diagram (ERD)

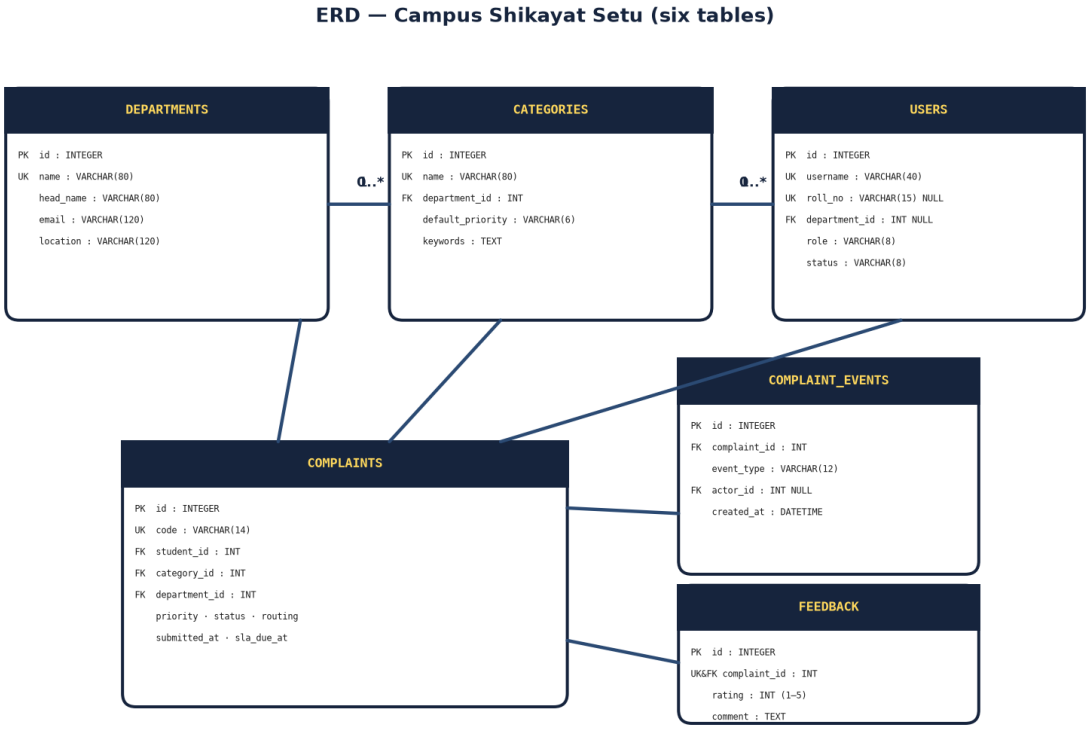


Figure 3.1 ERD — six tables with PK/FK crow's-foot cardinalities.

3.2.1 Data dictionary (summary)

Table	Key columns & constraints
departments	id PK · name UNIQUE · head_name · email · location
categories	id PK · name UNIQUE · department_id FK→departments · default_priority CHECK(High,Medium,Low) · keywords TEXT
users	id PK · role CHECK(student,officer,admin) · username UNIQUE · password_hash · roll_no UNIQUE NULL · department_id FK NULL · status CHECK(Active,Inactive)
complaints	id PK · code UNIQUE (CMP-YYYY-NNNN) · student_id FK · category_id FK · department_id FK · priority CHECK · routing CHECK(auto,manual) · routing_score · status CHECK · is_anonymous · submitted_at · sla_due_at · resolved_at
complaint_events	id PK · complaint_id FK ON DELETE CASCADE · event_type · note · actor_id FK NULL · created_at
feedback	id PK · complaint_id FK UNIQUE · rating CHECK(1..5) · comment · created_at

3.3 Use-Case Model

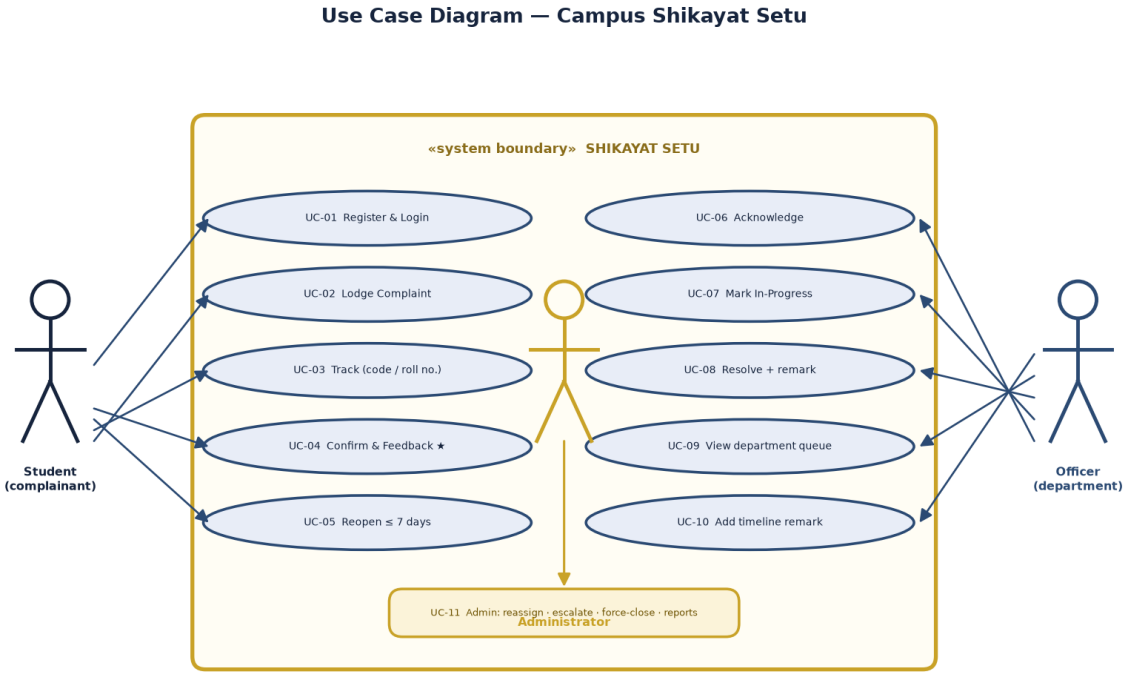


Figure 3.2 Use-case diagram — Student / Officer / Admin swimlanes with shared use cases.

3.3.1 Use-case descriptions (selection)

Use case	Actor	Flow
Lodge complaint	Student	Fill title+description+location+category → submit → router assigns dept & SLA → code shown on timeline.
Resolve complaint	Officer	Open queue → acknowledge → in-progress → resolve with remark → status Resolved; resolved_at = now.
Confirm & close	Student	Resolved list → star rating + comment → status Closed; feedback row inserted (UNIQUE per complaint).
Reopen (≤7 days)	Student	Resolved list → reason → status back to In Progress; event reopened logged.
Reassign	Admin	Console → pick new department → routing=manual → audit event with old→new.
Escalate to High	Admin	Console → escalate → priority=High → audit event; SLA recomputed? (kept original deadline; escalation visibility only).
Track without login	Public	Enter code + roll no. → timeline & current status rendered.

3.4 Activity Model — Lifecycle

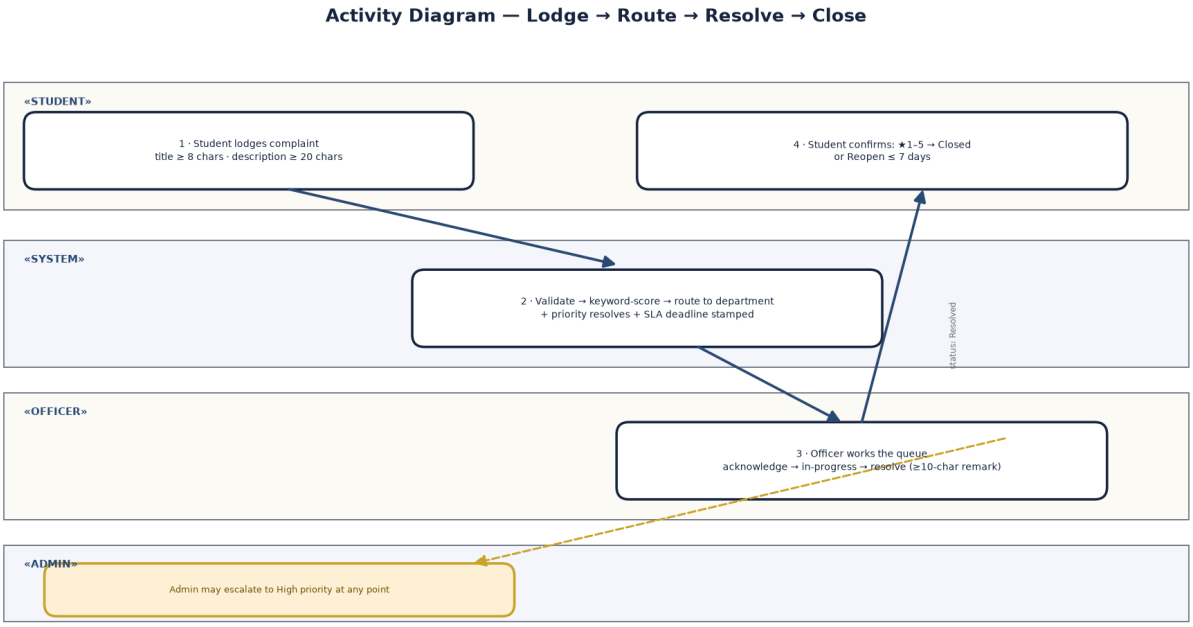


Figure 3.3 Activity diagram — lodging, routing, SLA stamping, officer workflow, closure path.

Overdue is never stored — it is derived: now > sla_due_at AND status NOT IN (Resolved, Closed). This keeps every register consistent even if the server is restarted after a long break.

3.5 Class Diagram

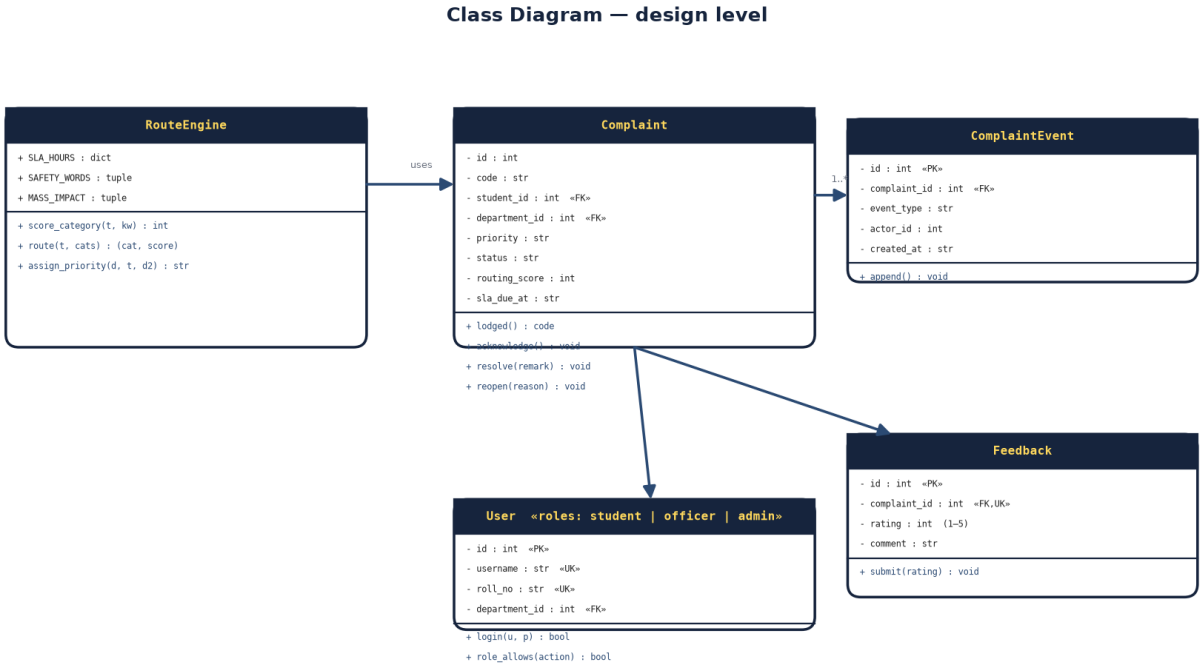


Figure 3.4 Class view — RouteEngine, Complaint, ComplaintEvent, Feedback, User hierarchy.

3.6 Deployment View

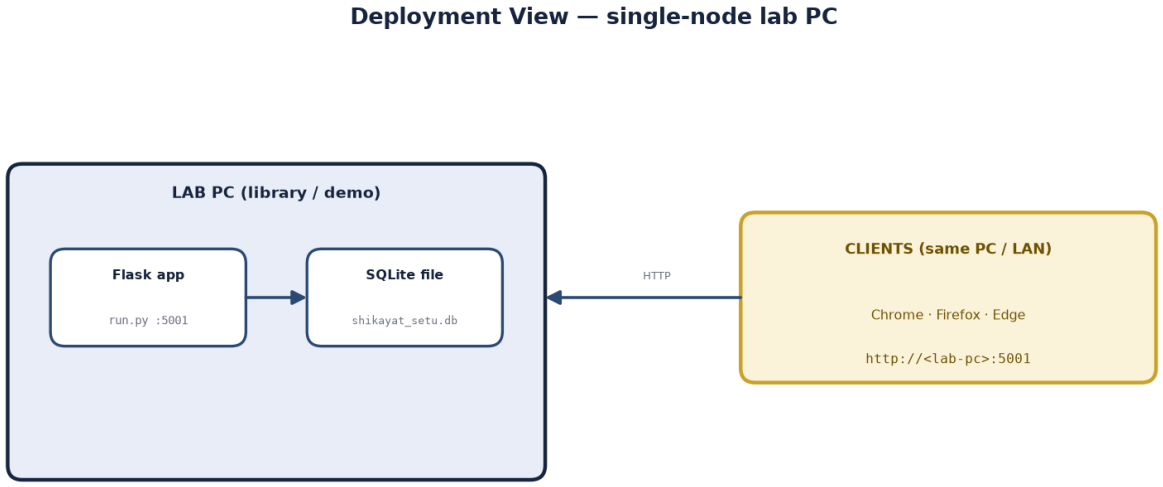


Figure 3.5 Deployment — single node lab PC; browsers on same PC or LAN peers.

3.7 Module Hierarchy

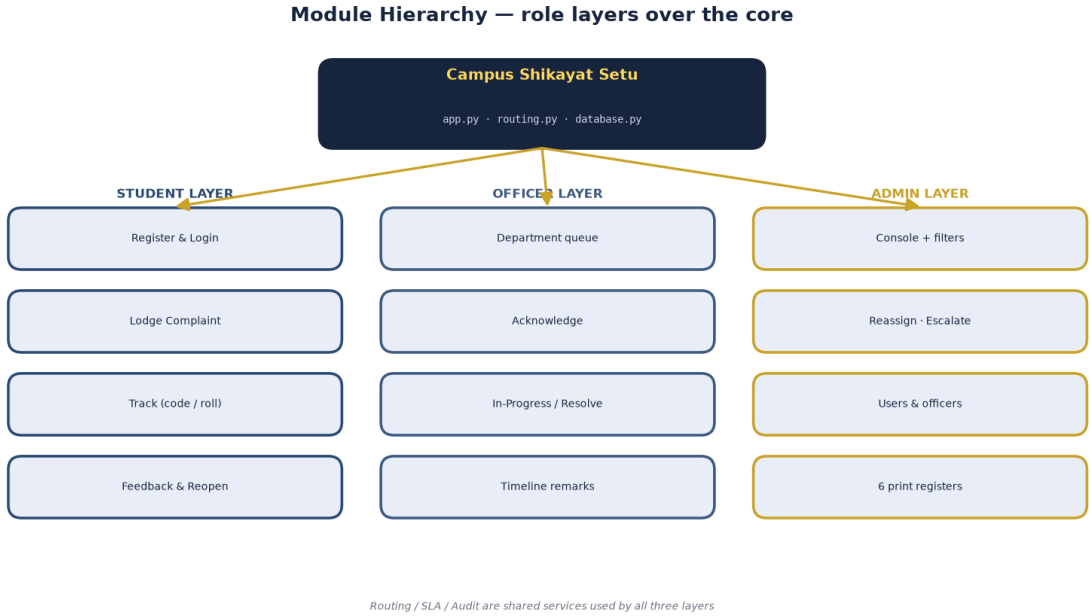


Figure 3.6 Module hierarchy — three role layers over the shared routing/SLA core.

3.8 Constraints and Assumptions

- One campus, one server process, SQLite file in the app directory; no concurrent-write contention in the lab.
- Keyword lists are curated per domain; sarcasm/vindication requires admin reassignment (Chapter 6, L-2).
- Complaint codes are sequential per calendar year; no deletion of complaints (audit-preserving design).
- Anonymous complaints are visible in full only to administrators; evening demo uses seeded admin account.

CHAPTER 4

Coding

Full source is in the repository under **shikayat-setu/**. Key excerpts below are verbatim from the running build (commit **527670a**).

4.1 Routing engine — scoring and SLA

routing.py — keyword scorer & priority resolver

```
def score_category(text: str, keywords: str) -> int:
    score = 0
    low = text.lower()
    for kw in (k.strip().lower() for k in keywords.split(",") if k.strip()):
        pattern = r"(?<![a-z0-9])" + re.escape(kw) + r"(?![a-z0-9])"
        hits = len(re.findall(pattern, low))
        if hits:
            weight = max(1, len(kw.split()) * 2)   # multi-word kws weigh more
            score += weight * hits
    return score

def assign_priority(default_priority, title, description):
    text = f"{title}. {description}".lower()
    if any(w in text for w in SAFETY_WORDS):
        return "High"
    p = default_priority
    if any(p_ in text for p_ in MASS_IMPACT):
        p = bump_priority(p)
    return p
```

4.2 Lodging a complaint — validation, routing, SLA stamping

app.py — lodge() excerpt (issue: placeholder-then-update to satisfy NOT NULL code)

```
title = (request.form.get("title") or "").strip()
desc  = (request.form.get("description") or "").strip()
if len(title) < 8:  errors.append("Title must be at least 8 characters.")
if len(desc)  < 20: errors.append("Description must be at least 20 characters.")

text = routing.route_text(title, desc)
routed_cat, score = routing.route(text, cats)
final_cat = routed_cat if (routed_cat and score > routing.score_category(text, chosen["keywords"])) else chosen
priority  = routing.assign_priority(final_cat["default_priority"], title, desc)
due       = now + timedelta(hours=routing.SLA_HOURS[priority])

cur = conn.execute("INSERT INTO complaints (code, student_id, ...) "
                   "VALUES ('TMP',?,?,?,?,?,?,?,?,'Submitted',?,?,?)", ...)
conn.execute("UPDATE complaints SET code=? WHERE id=?",
             (f"CMP-{now.year}-{cur.lastrowid:04d}", cur.lastrowid))
```

4.3 Resolution workflow — status transitions

app.py — resolve() excerpt

```
@app.route("/complaint/<code>/resolve", methods=["POST"])
@login_required
@role_required("officer", "admin")
def resolve(code):
    remark = (request.form.get("remark") or "").strip()
    ...
    if c["status"] not in ("In Progress", "Acknowledged", "Submitted"):
        flash("Complaint is already resolved or closed.", "warning")
    else:
        conn.execute("UPDATE complaints SET status='Resolved', resolved_at=? WHERE id=?",
                     (datetime.now().strftime("%Y-%m-%d %H:%M:%S"), c["id"]))
        log_event(conn, c["id"], "status", f"Resolved — {remark}", user["id"])
        conn.commit()
```

4.4 SLA compliance report — SQL

app.py — reports() SLA compliance query

```
SELECT d.name AS dept,
       COUNT(c.id) AS resolved_total,
       SUM(CASE WHEN c.resolved_at IS NOT NULL
                 AND c.resolved_at <= c.sla_due_at THEN 1 ELSE 0 END) AS within_sla,
       ROUND(100.0 * SUM(CASE WHEN c.resolved_at IS NOT NULL
                              AND c.resolved_at <= c.sla_due_at THEN 1 ELSE 0 END)
             / NULLIF(COUNT(c.id),0), 1) AS pct
FROM departments d
LEFT JOIN complaints c ON c.department_id=d.id AND c.status IN ('Resolved','Closed')
GROUP BY d.id ORDER BY pct DESC NULLS LAST;
```

4.5 Sample of schema constraints in force

schema.sql — CHECK constraints (integrity at the database layer)

```
status   TEXT NOT NULL DEFAULT 'Submitted'
         CHECK (status IN ('Submitted','Acknowledged','In Progress','Resolved','Closed')),
routing  TEXT NOT NULL DEFAULT 'auto' CHECK (routing IN ('auto','manual')),
priority TEXT NOT NULL CHECK (priority IN ('High','Medium','Low')),
rating   INTEGER NOT NULL CHECK (rating BETWEEN 1 AND 5)  -- feedback
```

CHAPTER 5

Testing

5.1 Testing Strategy

Two tracks were run: (A) automated tests via pytest (isolated temp database, seeded fresh per test), and (B) manual end-to-end scenarios on the live seeded app at http://127.0.0.1:5001. Both are reproducible from a clean clone.

5.2 Track A — Automated tests (12/12 passing)

ID	Scenario	Result
TC-01	Login page renders (200, brand present)	PASS
TC-02	Wrong password → HTTP 401	PASS
TC-03	Anonymous → 302 redirect to /login	PASS
TC-04	Admin dashboard renders overview KPIs	PASS
TC-05	Student self-registration happy path (MCA2026999)	PASS
TC-06	Duplicate roll number → 400 + error flash	PASS
TC-07	“leaking tap … washroom” auto-routes to Sanitation & Maintenance	PASS
TC-08	Safety keyword (“shock”) forces priority High	PASS
TC-09	Officer queue shows only own department; welfare complaint hidden from IT officer	PASS
TC-10	Full cycle lodge → acknowledge → in-progress → resolve → 5★ feedback → Closed	PASS
TC-11	Student forbidden from /console (HTTP 403)	PASS
TC-12	All six registers render for admin (200)	PASS

Command: cd shikayat-setu && python3 -m pytest tests/ -q
Output: 12 passed in ~16 s (isolated $TMPDIR database, seeded deterministically).

5.3 Track B — Manual scenarios (live, 09-11-2026)

ID	Scenario	Expected	Actual
TC-13	Login page in browser	Brand + demo credentials visible	PASS
TC-14	Admin login → dashboard KPIs (32 complaints)	Counts match seed	PASS
TC-15	Officer “it.officer” opens queue	Only IT & Computer Labs rows; ragging complaint absent	PASS
TC-16	Anonymous complaint CMP-2026-0015 viewed by officer	Identity shows “Anonymous”; admin sees full identity	PASS
TC-17	Lodge “fan regulator … classroom 3-A” as student	Routed to Sanitation & Maintenance, Medium, SLA 72 h	PASS
TC-18	Reopen resolved complaint within 7 days	Status returns to In Progress; event logged	PASS
TC-19	Public tracker: CMP-2026-0001 + roll MCA2026002	Timeline renders without login	PASS
TC-20	Ctrl+P on /reports?report=sla	Chrome hidden; table fits A4 portrait	PASS

5.4 Defect log (all found & fixed during the build)

ID	Defect	Fix	Verified
DEF-01	Seed crashed: complaints.code NOT NULL constraint on insert	Insert placeholder TMP-<hex>, then UPDATE to CMP-YYYY-NNNN using lastrowid	Re-seeded clean 32/32
DEF-02	Seed crashed: minute overflow (7*hour+3 could be ≥ 60)	minute = (13*hour + 7) % 60	Re-seeded clean
DEF-03	Officer dashboard passed sqlite Row to template expecting dict key is_overdue	Rows converted to dict with computed is_overdue before render	12/12 tests + manual TC-17

5.5 Regression

After each fix the full suite was re-run (12/12) plus the affected manual scenario; final state on 09-11-2026: **12 passed in 15.85 s; manual TC-13 … TC-20 all PASS.**

CHAPTER 6

Limitations of Proposed System

- **L-1 Explainable but dumb router:** the keyword engine cannot understand context or sarcasm; a mis-route needs admin reassignment (one click) — this is a feature for auditability, a limitation for raw accuracy.
- **L-2 Anonymous follow-up paradox:** sensitive complaints (e.g. ragging) hide identity from officers, so the officer cannot ask clarifying questions; only the admin can see the complainant.
- **L-3 No notifications:** students must poll /my-complaints or the tracker; no SMS/e-mail push yet.
- **L-4 Single node:** SQLite file on one PC is the whole database; horizontal scaling is impossible without migration.
- **L-5 Category catalogue is seed-fixed:** officers cannot add categories; admins cannot edit them without SQL.
- **L-6 Reopen window fixed at 7 days:** not configurable per department.
- **L-7 Reports are print-tables, not charts:** powerful for the register, weak for trend visualisation.
- **L-8 No pagination:** lists render fully; fine at ≤ 10k rows, a must-fix at 100k.
- **L-9 Single-semester evidence:** behaviour is validated on a seeded dataset + 12 tests, not on a live semester of usage.

Acceptance for the Review 3 build rests on three facts: (i) the complete lifecycle lodge → resolve → feedback → closed is executable and audit-trailed end to end; (ii) routing, SLA and escalation are deterministic and explainable on the timeline; (iii) the six registers reconcile to the seeded database row-for-row. Every limitation above maps to an enhancement in Chapter 7.

CHAPTER 7

Proposed Enhancements

ID	Enhancement	Effort	Addresses
E-1	E-mail + SMS on every status change (submission, resolve) via SMTP/SMS gateway	2 wks	L-3
E-2	WhatsApp deep-link tracker: share CMP code as wa.me link with prefilled text	3 days	L-3
E-3	Photo evidence upload (trap: validate MIME + size ≤ 2 MB; store under instance/)	1 wk	L-1
E-4	Admin category management UI (add/retire keywords, re-map departments)	1 wk	L-5
E-5	MySQL 8 migration; schema ports 1:1 (AUTOINCREMENT→AUTO_INCREMENT)	3 days	L-4
E-6	Nightly SLA-breach cron: auto-escalate to admin at 50%/100% of deadline	2 days	L-1, SLA
E-7	Charts dashboard (matplotlib PNGs or Chart.js): pendency, trend, SLA %	2 wks	L-7
E-8	Public board in the admin corridor: live counts, longest-pending grievances	1 wk	visibility
E-9	Marathi / Hindi UI toggle (Jinja-gettext patterns; 120 keys)	2 wks	accessibility
E-10	ERP bridge: sync roll no. ↔ ERP student master; auto e-mail to parents on High	3 wks	reach

Sequencing: E-3, E-4, E-6 are drop-in (no schema change). E-5 is the only migration and is deliberately trivial — the schema was written for it. E-8 gives the college an immediate visible win; E-10 is a research track, scoped but not promised.

CHAPTER 8

Conclusion

Campus Shikayat Setu replaces an untraceable complaint box with an auditable, SLA-bound grievance pipeline — built, seeded, tested and documented inside one semester. Every functional requirement FR-1…FR-12 is executable on the live build; every manual scenario TC-13…TC-20 passes; all known defects were fixed and re-verified.

The engineering lesson is old but worth repeating: for a single-campus grievance tool, a boring stack (Flask + SQLite + Jinja) outperforms any fashionable one — the database choice was made for the use case, the architecture is one file deep, and the demo is deterministic. The craft is in the rules on the timeline: routing score, SLA stamp, escalation, reopen window — small, explainable algorithms that turn anger about a broken tap into a closed-loop service.

The system is ready for Review 3 (21-11-2026) and for continued hardening toward the Final Internal Viva (19-12-2026), with the Chapter 7 roadmap as the post-viva backlog.

CHAPTER 9

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CHAPTER 10

User Manual

10.1 Roles & demo credentials

Role	Credentials	Can do
Admin	admin / admin123	Everything: console, reassign, escalate, force-close, users & officers, reports
Officer	hostel.officer · it.officer · maint.officer · exam.officer · welfare.officer · admin.officer / officer123	Own-department queue; acknowledge / progress / resolve; remarks
Student	MCA2026001 … MCA2026008 / student123	Lodge, track, feedback, reopen (≤ 7 days)
Public (no login)	/track + complaint code + roll no.	View timeline only

10.2 Quick start (on the demo PC)

run in five commands

```
git clone https://github.com/Yashwant-git392/Yashwant-git392.git
cd Yashwant-git392/shikayat-setu
python3 -m venv .venv && source .venv/bin/activate   # optional
pip install -r requirements.txt
python3 run.py        # → http://127.0.0.1:5001  (auto-initialises & seeds)
```

10.3 The five screens you will touch most

10.3.1 Sign in

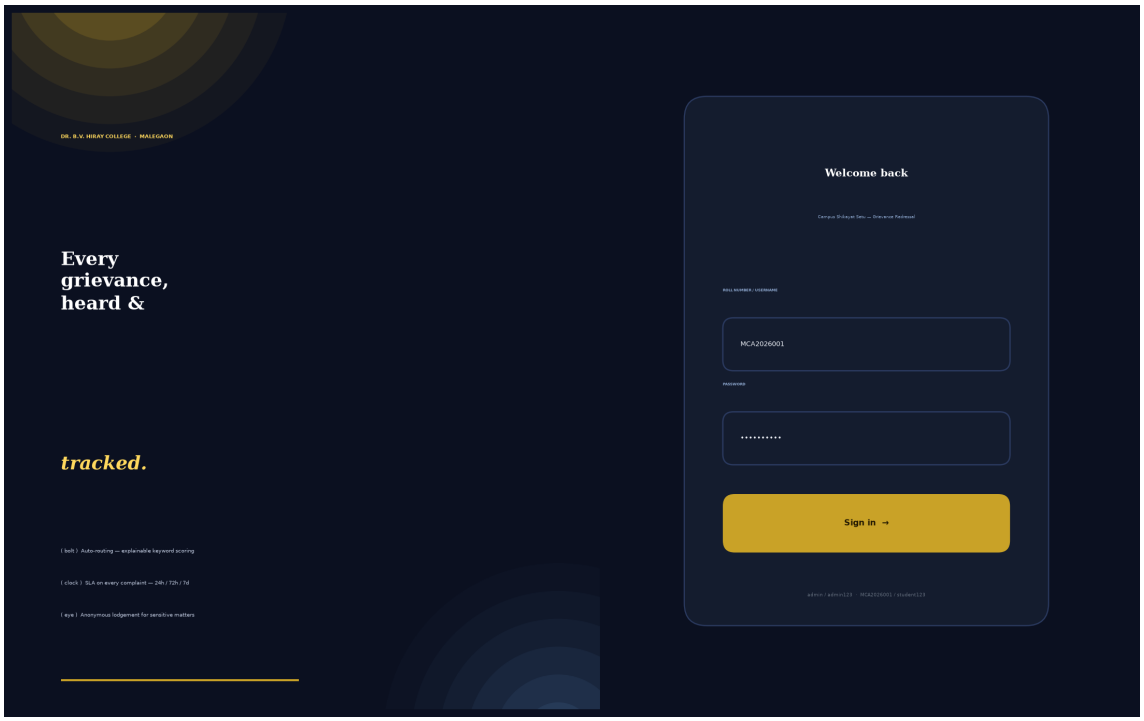


Figure 10.1 Sign-in — roll no. or username; demo credentials shown below the form.

10.3.2 Student dashboard — my complaints at a glance

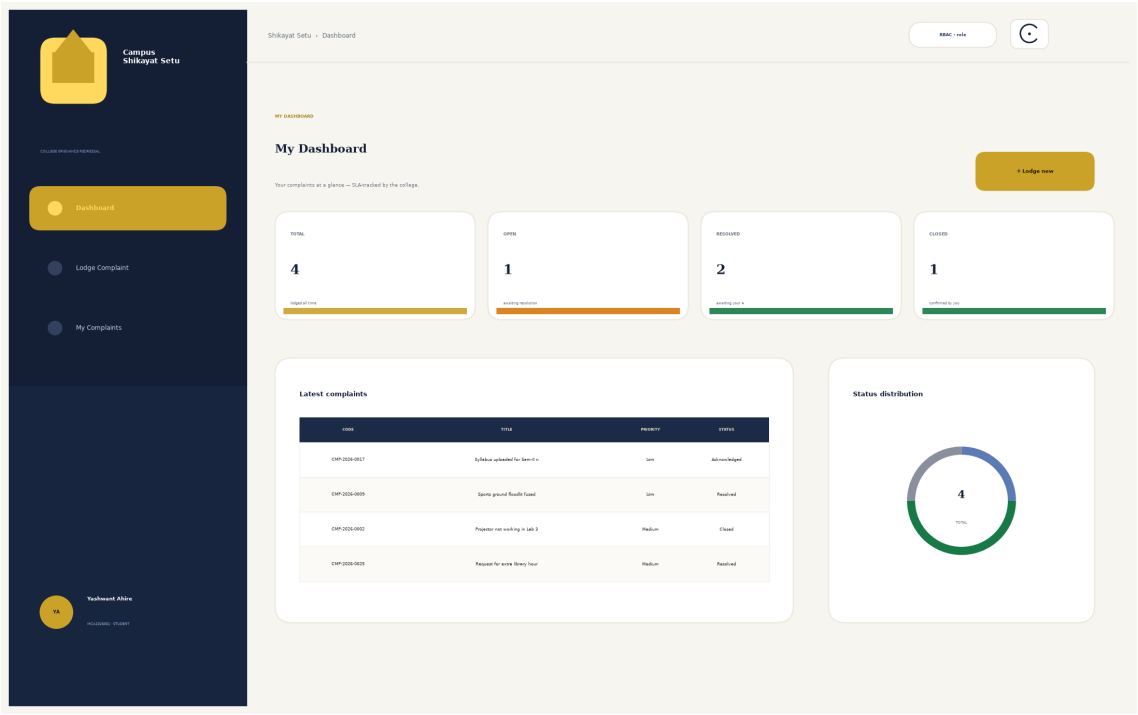


Figure 10.2 Student dashboard — KPI tiles (total, submitted…closed) and latest complaints.

10.3.3 Admin console — every complaint, filterable

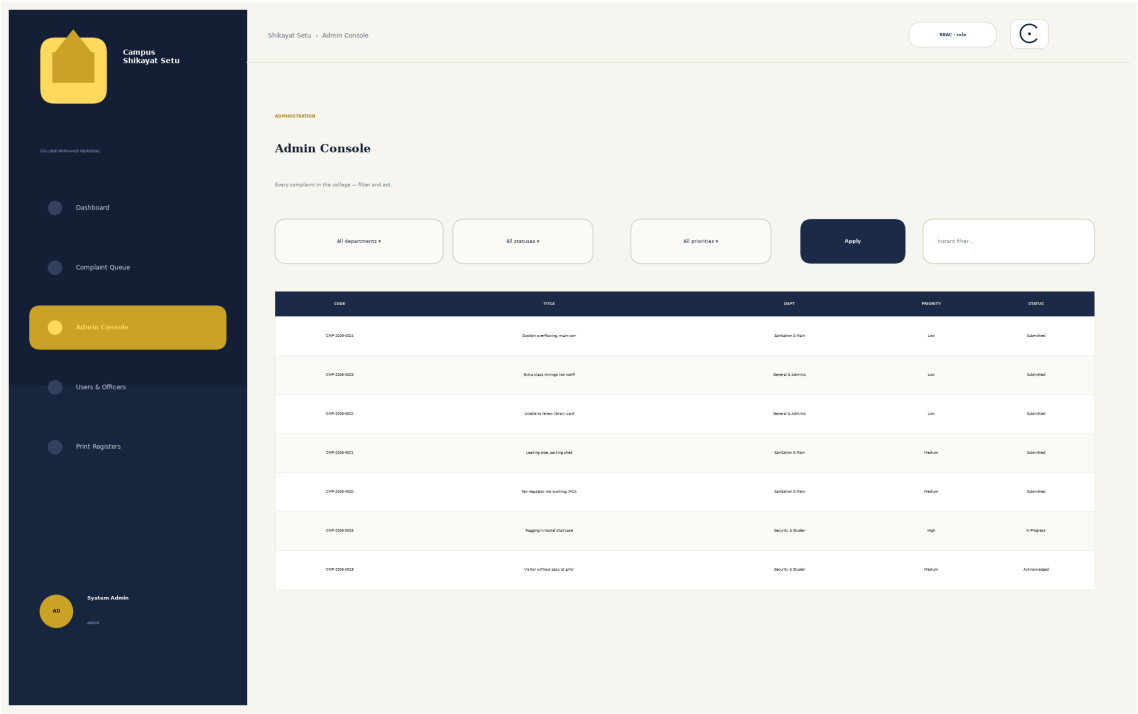


Figure 10.3 Admin console — department/status/priority filters; escalate, reassign.

10.3.4 Complaint detail — the timeline is the audit trail

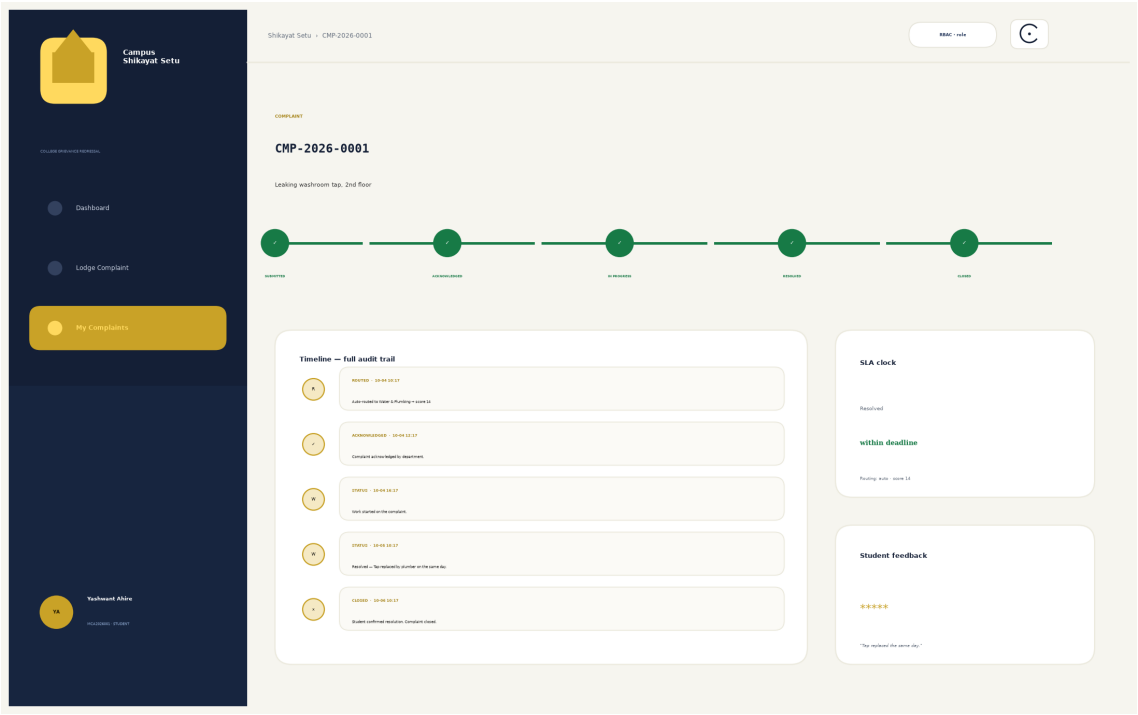


Figure 10.4 Complaint detail — SLA badge, routing score, full event timeline.

10.4 Frequently asked questions

- **“The complaint is routed wrong!”** — Admins reassign in one click; the timeline records old → new department.
- **“How do I see overdue complaints?”** — the reports tab “Department Pendency” has an Overdue column; officer queue sorts High priority first.
- **“Can I edit a complaint after lodging?”** — Not yet; add a remark as officer, or reopen as student. Deliberate: audit integrity.
- **“Anonymous complaint — who sees who I am?”** — Only the administrator. Officers see “anonymous”.
- **“Where is the data file?”** — app/shikayat_setu.db. Delete it and restart to re-seed the demo.
- **“Can it run on Windows?”** — Yes. Python 3.11 + two pip packages; no services.

APPENDIX A

Appendix A — Database Schema (schema.sql)

schema.sql (verbatim)

```
-- Campus Shikayat Setu -- College Grievance Redressal System
-- SPPU MCA-I (Sem-I) Application Development mini project
-- Creator: Yashwant Santosh Ahire
PRAGMA foreign_keys = ON;

CREATE TABLE IF NOT EXISTS departments (
    id          INTEGER PRIMARY KEY AUTOINCREMENT,
    name        TEXT NOT NULL UNIQUE,
    head_name   TEXT NOT NULL,
    email       TEXT NOT NULL,
    location    TEXT NOT NULL
);

CREATE TABLE IF NOT EXISTS categories (
    id               INTEGER PRIMARY KEY AUTOINCREMENT,
    name             TEXT NOT NULL UNIQUE,
    department_id    INTEGER NOT NULL REFERENCES departments(id) ON DELETE CASCADE,
    default_priority TEXT NOT NULL CHECK (default_priority IN ('High','Medium','Low')),
    keywords         TEXT NOT NULL
);

CREATE TABLE IF NOT EXISTS users (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    role          TEXT NOT NULL CHECK (role IN ('student','officer','admin')),
    username      TEXT NOT NULL UNIQUE,
    password_hash TEXT NOT NULL,
    full_name     TEXT NOT NULL,
    roll_no       TEXT UNIQUE,                      -- students only
    phone         TEXT,
    department_id INTEGER REFERENCES departments(id), -- officers only
    status        TEXT NOT NULL DEFAULT 'Active' CHECK (status IN ('Active','Inactive')),
    created_at    TEXT NOT NULL DEFAULT (datetime('now','localtime'))
);

CREATE TABLE IF NOT EXISTS complaints (
    id            INTEGER PRIMARY KEY AUTOINCREMENT,
    code          TEXT NOT NULL UNIQUE,             -- CMP-YYYY-NNNN
    student_id    INTEGER NOT NULL REFERENCES users(id),
    category_id   INTEGER NOT NULL REFERENCES categories(id),
    department_id INTEGER NOT NULL REFERENCES departments(id),
    title         TEXT NOT NULL,
    description   TEXT NOT NULL,
    location      TEXT NOT NULL DEFAULT '',
    priority      TEXT NOT NULL CHECK (priority IN ('High','Medium','Low')),
    routing       TEXT NOT NULL DEFAULT 'auto' CHECK (routing IN ('auto','manual')),
    routing_score INTEGER NOT NULL DEFAULT 0,
    status        TEXT NOT NULL DEFAULT 'Submitted'
                  CHECK (status IN ('Submitted','Acknowledged','In Progress','Resolved','Closed')),
    is_anonymous  INTEGER NOT NULL DEFAULT 0,
    submitted_at  TEXT NOT NULL DEFAULT (datetime('now','localtime')),
    sla_due_at    TEXT NOT NULL,
    resolved_at   TEXT
);
CREATE INDEX IF NOT EXISTS idx_complaints_dept_status ON complaints(department_id, status);
CREATE INDEX IF NOT EXISTS idx_complaints_student     ON complaints(student_id);

CREATE TABLE IF NOT EXISTS complaint_events (
    id           INTEGER PRIMARY KEY AUTOINCREMENT,
    complaint_id INTEGER NOT NULL REFERENCES complaints(id) ON DELETE CASCADE,
     event_type    TEXT NOT NULL,     -- routed | acknowledged | status | remark | reassign | escalated | feedback |
reopened | closed | created
```

```
    note         TEXT NOT NULL DEFAULT '',
    actor_id     INTEGER REFERENCES users(id),
    created_at   TEXT NOT NULL DEFAULT (datetime('now','localtime'))
);
CREATE INDEX IF NOT EXISTS idx_events_complaint ON complaint_events(complaint_id);

CREATE TABLE IF NOT EXISTS feedback (
    id           INTEGER PRIMARY KEY AUTOINCREMENT,
    complaint_id INTEGER NOT NULL UNIQUE REFERENCES complaints(id) ON DELETE CASCADE,
    rating       INTEGER NOT NULL CHECK (rating BETWEEN 1 AND 5),
    comment      TEXT NOT NULL DEFAULT '',
    created_at   TEXT NOT NULL DEFAULT (datetime('now','localtime'))
);
```

APPENDIX B

Appendix B — Seed Data (actual records)

B.1 departments — 6 rows

Name	Head	Email	Location
Hostel Office	Mrs. S. Kulkarni	hostel@bvhrcet.ac.in	Hostel Block A, Ground Floor
IT & Computer Labs	Mr. R. Pawar	itlabs@bvhrcet.ac.in	Computer Block, 2nd Floor
Sanitation & Maintenance	Mr. V. Gade	maintenance@bvhrcet.ac.in	Maintenance Office, Main Gate
Examination Cell	Mrs. A. Borse	examcell@bvhrcet.ac.in	Administrative Block, 1st Floor
Security & Student Welfare	Mr. D. Shinde	welfare@bvhrcet.ac.in	Main Building, Reception
General & Administration	Mrs. P. Jagtap	admin@bvhrcet.ac.in	Administrative Block, Ground Floor

B.2 categories — 15 rows (keyword router fodder)

Category	Department	Default	Keywords (seed)
Ragging & Harassment	Security & Student Welfare	High	ragging, harass, bullying, misbehave, threat, eve teasing, intimid
Electrical & Safety	Sanitation & Maintenance	Medium	fan, light, tube, wiring, switch, socket, electrical, shock, short circuit, power failure, voltage
Water & Plumbing	Sanitation & Maintenance	Medium	leak, pipe, tap, water, drain, toilet, washroom, bathroom, flush, sewage, tap leaking
IT & Network	IT & Computer Labs	Medium	computer, pc, wifi, network, internet, projector, printer, lab, software, login, portal, server, printer not
Hostel Room & Mess	Hostel Office	Medium	hostel, room, bed, mess, food, canteen food, curfew, outpass, roommate, blanket, hostel mess, geyser
Classroom Facility	Sanitation & Maintenance	Low	bench, desk, chair, classroom, board, chalk, door, window, sweeper, cleaning, dustbin
Examination & Marks	Examination Cell	High	exam, marks, admit card, hall ticket, result, grade, revaluation, rechecking, certificate, transcript, degree
Fees & Finance	Examination Cell	Medium	fees, receipt, scholarship, payment, concession, fine receipt, challan
Library & Reading Room	General & Administration	Low	library, book issue, book return, reading room, library card, overdue
Academics & Faculty	General & Administration	Low	syllabus, lecture, faculty, professor, notes, assignment, class cancellation, timetable
Sports & Activities	Security & Student Welfare	Low	sports, ground, equipment, tournament, gym, playground
Security & Gate Pass	Security & Student Welfare	Medium	gate, security, entry, visitor, gate pass, theft, lost, stolen
Health & Canteen	Hostel Office	High	food poisoning, canteen, hygiene, medicine, sick, injury, first aid, sick room, canteen food spoiled
Canteen & Food	General & Administration	Medium	canteen, food quality, snacks, canteen price, tea, menu
General Query / Other	General & Administration	Low	general, other, query, suggestion, feedback, information, help

B.3 users — 15 rows (admin, 6 officers, 8 students)

Username	Role	Full name	Department	Status
admin	admin	System Administrator	—	Active
admin.officer	officer	Mrs. P. Jagtap	General & Administration	Active
exam.officer	officer	Mrs. A. Borse	Examination Cell	Active
hostel.officer	officer	Mrs. S. Kulkarni	Hostel Office	Active
it.officer	officer	Mr. R. Pawar	IT & Computer Labs	Active
maint.officer	officer	Mr. V. Gade	Sanitation & Maintenance	Active
welfare.officer	officer	Mr. D. Shinde	Security & Student Welfare	Active
MCA2026001	student	Yashwant Santosh Ahire	—	Active
MCA2026002	student	Priya Patil	—	Active
MCA2026003	student	Rohit Deshmukh	—	Active
MCA2026004	student	Snehal Jadhav	—	Active
MCA2026005	student	Aishwarya Gaikwad	—	Active
MCA2026006	student	Suraj Pawar	—	Active
MCA2026007	student	Vaishnavi Kulkarni	—	Active
MCA2026008	student	Tejas Waghmare	—	Active

B.4 complaints — 32 rows (₹-style totals not applicable)

Code	Title	Pri	Status
CMP-2026-0001	Leaking washroom tap, 2nd floor	Medium	Closed
CMP-2026-0002	Projector not working in Lab 3	Medium	Closed
CMP-2026-0003	Tube light fused in classroom 2-B	Medium	Closed
CMP-2026-0004	Marks not shown in ERP for Paper-III	High	Closed

Code	Title	Pri	Status
CMP-2026-0005	Bench broken in classroom 1-C	Low	Closed
CMP-2026-0006	Mess food quality poor this week	High	Resolved
CMP-2026-0007	Gate pass lost near main gate	Medium	Resolved
CMP-2026-0008	Toilet flush not working, hostel 2nd floor	Medium	Resolved
CMP-2026-0009	Sports ground floodlit fused	Low	Resolved
CMP-2026-0010	Corridor lights fused on ground floor	Medium	In Progress
CMP-2026-0011	Wi-Fi in library wing not reachable	Medium	In Progress
CMP-2026-0012	Water geyser not heating, room 214	Medium	In Progress
CMP-2026-0013	Revaluation result pending since June	High	In Progress
CMP-2026-0014	Canteen food hygiene complaint	High	In Progress
CMP-2026-0015	Ragging in hostel staircase	High	In Progress
CMP-2026-0016	Lab 1 software crash on opening ERP	High	Acknowledged
CMP-2026-0017	Syllabus uploaded for Sem-II not opening	Low	Acknowledged
CMP-2026-0018	Visitor without pass at girls' hostel	Medium	Acknowledged
CMP-2026-0019	Scholarship concession receipt not generated	Medium	Acknowledged
CMP-2026-0020	Fan regulator not working, MCA-I	Medium	Submitted
CMP-2026-0021	Leaking pipe, parking shed	Medium	Submitted
CMP-2026-0022	Unable to renew library card	Low	Submitted
CMP-2026-0023	Extra class timings not notified	Low	Submitted
CMP-2026-0024	Dustbin overflowing, main corridor	Low	Submitted
CMP-2026-0025	Request for extra library hours	Medium	Resolved
CMP-2026-0026	ID card correction on ERP	Medium	Resolved
CMP-2026-0027	Water cooler not cooling, near gym	Medium	In Progress
CMP-2026-0028	Volleyball posts loosely fitted	Low	Acknowledged
CMP-2026-0029	Fine receipt mismatch for hostel fees	Medium	Closed
CMP-2026-0030	ERP login failing for few students	Medium	Acknowledged
CMP-2026-0031	Sick room medicines expired	High	Resolved
CMP-2026-0032	Block outpass not issued for Sunday	Medium	Resolved

B.5 complaint_events — 100 rows (audit trail)

Event type	Count
status	35
routed	32
acknowledged	27
closed	6

B.6 feedback — 6 rows

Complaint	Rating	Comment
CMP-2026-0001	★★★★★	Fixed quickly, thank you.
CMP-2026-0002	★★★★★	Perfect.
CMP-2026-0003	★★★★	Good but took one extra day.
CMP-2026-0004	★★★★★	Resolved.
CMP-2026-0005	★★★★	—
CMP-2026-0029	★★★★★	Rectified, thank you.

APPENDIX C

Appendix C — Glossary and Acronyms

Term	Meaning
Audit event	Immutable row in complaint_events recording every state change with actor and timestamp.
Auto-routing	Keyword-weighted assignment of a complaint to a department; scored and printed on the timeline.
CMP code	Human-readable complaint identifier: CMP-<year>-<4-digit sequence>.
Force-close	Admin-only closure of a Resolved complaint after a 7-day no-response window.
Grievance	Any student-raised deficiency in college services — facility, academic, security, food.
Keyword weight	Routing score = Σ(weight × hits); multi-word keywords weigh double per word.
Mass impact	Phrase class (“entire”, “all students”) that bumps priority one level.
OLA	Operational Level Agreement — internal equivalent of SLA between departments.
Reopen	Student rejection of a resolution within 7 days; returns status to In Progress.
Resolution remark	Mandatory ≥ 10-character note officers must enter when resolving.
Safety word	Phrase class (ragging, shock, fire...) that forces priority High at submission.
SLA	Service-Level Agreement — High 24 h · Medium 72 h · Low 7 days for this college.
Timeline	Chronological list of a complaint’s audit events; the single source of truth.